

Banana Pi M4Zero A1 Porting Correction and DDR Tuning

A focused 792 MHz correction, observed margin evidence, standard image delivery, and the path to qualification

STATUS | ENGINEERING VALIDATION CANDIDATE

Document control	Value
Document revision	1.0
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Target platform	Banana Pi M4Zero A1
DDR target	792 MHz
Standard customer validation build	P02e5
Intended audience	Customers, technical partners, product managers, and customer engineering teams

This report summarizes verified repository evidence available at the document date. It defines an engineering validation candidate. It does not constitute production qualification or a stable release.

DECISION BRIEF

Executive Summary

Outcome

The Banana Pi M4Zero A1 correction retains the 792 MHz DDR target and the established `TPR6=0x3a808080` center setting. It makes a minimal paired lane update: `TPR11` changes from `0x24242422` to `0x25252523`, and `TPR12` changes from `0x110f1111` to `0x110f0f10`.

The correction was required because the previous 792 MHz lane parameter pairing showed insufficient margin on an additional hardware sample under the stronger M2 data-integrity test. One of five repetitions produced a data mismatch. This evidence identifies a margin limitation in that parameter pairing under the tested condition. It does not support blaming the root filesystem, and it does not establish a unique physical root cause.

Verified results at a glance

Area	Verified result	Customer interpretation
Parameter correction	792 MHz and <code>TPR6=0x3a808080</code> retained; <code>TPR11</code> and <code>TPR12</code> updated as a pair	The change is narrow and traceable
Observed <code>TPR6</code> window	Upper-byte passing window <code>0x30..0x42</code> ; deliberate failure boundaries <code>0x2e</code> and <code>0x44</code>	Useful engineering margin evidence, not a production tolerance guarantee
A1 center candidate	64 MiB M2 five-pass pattern testing <code>20/20</code> ; safe recovery <code>20/20</code>	No failure was observed in this defined validation sequence
Evidence dataset	322 records: 270 passes and 52 retained convergence or boundary negative results	The 52 negative results are not an A1 candidate failure rate
Standard Linux boot	Covered 2 GiB single-rank and 4 GiB dual-rank configurations	Boot-chain and memory-geometry coverage, not long-duration qualification
Image delivery	10 images across five distributions, each in CLI and XFCE variants	Checksums and image plus bootloader package identity were verified

Status statement. A1 at 792 MHz is an engineering validation candidate. It is not production-qualified and is not a stable release.

Recommended decision

Use standard build `P02e5` for controlled customer evaluation while the remaining qualification gates are completed. Keep experimental runtime scanning binaries separate from the standard bootloader throughout preparation, test, and feedback collection.

CORRECTION RATIONALE

1. Why the A1 Correction Was Needed

Stronger testing exposed a real margin limitation

Earlier checks did not expose a repeatable data-integrity problem at the target frequency. The stronger M2 method expanded pattern and address coverage, including coverage intended to exercise more demanding memory regions. Under that method, the previous 792 MHz lane pairing completed four of five repetitions and produced one explicit data mismatch.

That result changed the engineering interpretation. A boot symptom alone can have several causes, but a data mismatch under a controlled DDR test is direct evidence that the previous lane pairing did not provide sufficient observed margin on the additional sample. The available data supports a parameter-level correction. It does not isolate memory silicon, PCB routing, power delivery, temperature, or any other physical factor as the sole cause.

Minimal correction strategy

Parameter	Previous 792 MHz setting	A1 setting	Decision
DDR frequency	792 MHz	792 MHz	Retained
TPR6	0x3a808080	0x3a808080	Retained at the observed center candidate
TPR11	0x24242422	0x25252523	Updated
TPR12	0x110f1111	0x110f0f10	Updated as a pair with TPR11

The paired update was chosen because TPR11 and TPR12 encode related lane settings. Short comparison runs showed that changing TPR11 produced the larger observed improvement, while changing only TPR12 did not. The final engineering choice nevertheless updates and validates the two packed parameters together, avoiding an unsupported split configuration.

What remained intentionally unchanged

The frequency target, drive settings, termination settings, TPR6 , and TPR10 were not changed. Preserving those established values limited the number of variables and kept the correction focused on the parameter pair implicated by the additional sample evidence.

Engineering principle. The A1 correction is a controlled lane-parameter adjustment at the existing performance target, not a broad retuning of the DDR configuration.

MARGIN AND CONVERGENCE

2. DDR Window and Validation Evidence

Observed TPR6 upper-byte window

After applying the A1 lane pair, discrete M2 scans were used to observe the TPR6 upper-byte envelope around the retained center candidate.

Region or point	Observed result	Meaning
0x2e	Deliberate failure boundary	Nearest demonstrated lower-side failure point
0x30..0x42	Passing window at the tested points	Observed zero-failure engineering window
0x3a	Selected center candidate	Retained value in TPR6=0x3a808080
0x44	Deliberate failure boundary	Nearest demonstrated upper-side failure point

The 0x30..0x42 result is an observed engineering window from discrete points under the recorded test conditions. It is not a production tolerance guarantee, and it must not be extrapolated across all units, batches, temperatures, or supply conditions.

Center-candidate strength

The A1 center candidate completed a 64 MiB M2 sequence using a five-pass pattern test. The recorded outcome was:

Validation item	Result	Scope
64 MiB M2 five-pass pattern repetitions	20/20	A1 center candidate at 792 MHz
Return to the 480 MHz safe recovery setting	20/20	Recovery after each center-candidate sequence

These results demonstrate repeatability within the defined warm-reset engineering sequence. They do not replace complete power-cycle testing or extended Linux stress.

Correct interpretation of the 322-record dataset

Dataset class	Records	Interpretation
Pass	270	Passing records from coarse scans, pair comparisons, window work, and center validation
Retained negative result	52	Convergence evidence, including old-setting counter-evidence and deliberate boundary searches
Total	322	Full retained engineering dataset

The 52 retained negative results are essential to showing where settings stopped passing. They combine intentionally weak points, boundary exploration, and pre-correction evidence. Therefore, 52/322 is not an A1 center-candidate failure rate. The direct center-candidate result is the separate 20/20 result above.

STANDARD DELIVERY COVERAGE

3. Standard Boot and Image Matrix

Customer validation build identity

P02e5 is the standard customer validation build for A1 evaluation. It contains the fixed A1 DDR parameters and the standard boot path. A separate internal runtime scanning build was used to vary parameters during engineering convergence. That scanner is not the customer validation bootloader and must not be mixed into standard image testing.

Build role	Intended use	Customer action
Standard build P02e5	Normal A1 boot and controlled customer evaluation	Use this build and record it in every test result
Internal runtime scanning build	Engineering-only parameter exploration	Do not combine it with the standard bootloader or customer image results

Standard Linux boot coverage

Standard Linux boot records cover both represented memory geometry classes in the available sample set. The records include one 2 GiB single-rank sample and three 4 GiB dual-rank samples.

Memory configuration	Verified coverage	Limit
2 GiB single-rank	Geometry detection, standard boot chain, Linux user space, and orderly shutdown	Not a cold-boot count or long-duration stress result
4 GiB dual-rank	Geometry detection, standard boot chain, Linux user space, and orderly shutdown	Not a batch or environmental qualification result

The records demonstrate that the same A1 parameter set reached standard Linux user space across both memory configurations. The source boot evidence used a Noble user space. It did not bind every boot record to a specific CLI or XFCE image file.

Ten-image delivery matrix

Distribution	CLI	XFCE
Bookworm	Included	Included
Jammy	Included	Included
Noble	Included	Included
Resolute	Included	Included
Trixie	Included	Included

For all 10 delivered image variants, the retained verification established compressed XZ checksums, decompressed raw IMG checksums, embedded P02e5 identity, and bootloader-region identity. The standard bootloader package identity was also verified. These checks confirm delivery integrity and traceability. They do not mean that all 10 images completed full hardware stress qualification.

Customer checkpoint. Preserve the supplied checksum manifests with each image. After writing media, perform the readback gate before treating a file checksum as proof of the flashed content.

CLAIMS AND QUALIFICATION

4. What the Evidence Demonstrates

Claims and limitations

Evidence-backed claim	What is demonstrated	What is not demonstrated
A1 parameter correction	The defined engineering sequence passed after a narrow paired lane update	A unique physical root cause
Observed TPR6 window	Passing points from 0x30..0x42 with failures at 0x2e and 0x44	A production tolerance, batch-wide, or environmental guarantee
Center validation	64 MiB M2 five-pass pattern testing 20/20 and safe recovery 20/20	Lifetime reliability, controlled cold-boot statistics, or extended operating stress
Memory-geometry boot coverage	Standard Linux boot on 2 GiB single-rank and 4 GiB dual-rank configurations	Broader lot coverage or all operating corners
Ten-image matrix	Image checksums, embedded build identity, and bootloader package identity for five distributions in CLI and XFCE forms	Full hardware stress completion for every image

Remaining qualification gates

Gate	Required work	Completion evidence
Controlled complete power-cycle cold boot	Run frozen off-time and on-time sequences on representative memory configurations	Complete per-cycle UART records and pass/fail accounting
Flashed-media and bootloader readback	Verify the written image range and the bootloader region before first boot	Readback hashes matched to the exact supplied image
Long-duration concurrent stress	Run Linux memory testing together with CPU and storage load for a predefined duration	Commands, duration, load, temperature, exit status, and error review
Broader common-window coverage	Expand across batches, memory geometries, weaker units, temperature, and supply corners	A shared passing intersection with predefined acceptance rules

All four gates remain open. Until they are closed with controlled evidence, A1 remains an engineering validation candidate.

Qualification boundary. Standard boot success and a strong center-candidate result are necessary evidence, but they are not substitutes for cold boot, media readback, concurrent stress, and broader common-window coverage.

CUSTOMER EVALUATION

5. Evaluation and Feedback Guidance

Recommended evaluation sequence

1. Use the standard customer validation build `P02e5` without substituting an experimental scanner binary.
2. Preserve the original compressed-image checksum and the decompressed raw-image checksum with the test record.
3. Record the board memory geometry, including capacity and single-rank or dual-rank detection.
4. Write the selected image, then complete flashed-media and bootloader readback before the first validation boot.
5. Capture the complete UART log from initial power application through Linux shutdown or the point of failure.
6. Record the exact image identity: distribution, CLI or XFCE variant, full filename, and supplied checksum.
7. Keep power-cycle conditions, test commands, duration, and acceptance criteria fixed within a comparison set.

Feedback package

For efficient joint analysis, each customer result should include:

Field	Required detail
Build	<code>P02e5</code>
Image	Exact distribution, variant, filename, and checksum
Memory	Reported capacity and rank geometry
Media	Write method and readback checksum result
Power sequence	Controlled off-time, on-time, and cycle number
Test	Exact command, duration, load combination, and exit status
Logs	Complete UART output, including the earliest boot text and any failure context
Environment	Supply setup and available temperature information

Do not merge scanner output with standard boot results. If experimental runtime scanning is required, treat it as a separate engineering activity with separately identified binaries and records.

Conclusion and recommended next steps

The A1 correction is technically credible, deliberately small, and supported by observed margin boundaries, a `20/20` center-candidate result, standard boot coverage across both memory geometries, and a traceable ten-image delivery matrix. The appropriate next step is controlled customer validation with `P02e5`, beginning with power-cycle cold boot and media readback, followed by concurrent Linux stress and broader common-window work.

Final status. Proceed with controlled engineering evaluation. Do not promote A1 beyond engineering validation candidate status until the remaining qualification gates are complete.

TECHNICAL REFERENCE

Appendix A. Parameter Delta and Validation Summary

A.1 Parameter delta

Item	Previous setting	A1 setting	Change status
DDR clock	792 MHz	792 MHz	Retained
TPR6	0x3a808080	0x3a808080	Retained
TPR11	0x24242422	0x25252523	Updated
TPR12	0x110f1111	0x110f0f10	Updated

A.2 Validation summary

Validation item	Verified result	Qualification note
Previous lane pair at 792 MHz under M2	4/5, including one data mismatch	Established insufficient observed margin on the additional sample
A1 observed passing window	0x30 . . 0x42	Discrete engineering window only
Deliberate failure boundaries	0x2e and 0x44	Demonstrated lower-side and upper-side failures
A1 center candidate	64 MiB M2 five-pass pattern 20/20	Defined warm-reset engineering sequence
Safe recovery	20/20	Returned to 480 MHz safe setting after each sequence
Full convergence dataset	322 records: 270 pass, 52 retained negative	Negative results are not an A1 candidate failure rate
Standard boot geometry	2 GiB single-rank and 4 GiB dual-rank	Standard Linux boot coverage only
Image matrix	10/10 identities verified	Five distributions, each CLI and XFCE; hardware stress remains selective

A.3 Document control

Control field	Value
Revision	1.0
Date	20 August 2026
Standard customer validation build	P02e5
Status	Engineering validation candidate
Source basis	Verified engineering report and repository evidence available at the report date

This document is intentionally concise. Detailed internal scan logs, hardware identifiers, workstation locations, and development history are excluded. The stated results should be interpreted only within the scopes and limitations shown in this report.